II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 16

BPH

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 148	Anatomy and Development of the Prostate
Ch 149	Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History
Ch 150	Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia
Ch 151	Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia
Ch 152	Simple Prostatectomy: Open, Laparoscopic and Robotic
★	Putting it together: the BPH pathway

148 Anatomy and Development of the Prostate

1 · Development (embryology)

The androgen pathways that build the prostate also drive BPH.

- **Origin**: the urogenital sinus (UGS), endoderm in mesoderm. Incomplete cloacal division (imperforate anus, urorectal fistula) impairs development.
- **Dihydrotestosterone (DHT)**: †5α-reductase type 2 in UGS mesenchyme converts testosterone to DHT, the more potent androgen receptor (AR) ligand. **SRD5A2** loss leaves the prostate incomplete; castration or inactive AR leaves it absent. †The same enzyme is the target of 5α-reductase inhibitors (5-ARIs) in BPH.
- **Androgen is not rate-limiting**: testosterone is present at week 9 but budding starts at week 10, gated by **AR demethylation** in the UGS.

Week	Event
6–7	Cloaca divides into the ventral UGS and the dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR, whose andromedins instruct the epithelium
10	Epithelial buds invade the mesenchyme; this fixes the adult ductal pattern

Mesenchyme instructs, epithelium responds

- Budding and branching need **mesenchymal (stromal)** AR, not epithelial AR; UGS mesenchyme can even turn bladder epithelium into prostate (Cunha).
- **Andromedins** are androgen-induced stromal signals, chiefly **fibroblast growth factor 10 (FGF10)**; Nkx3-1 specifies the epithelium.

> **KEY POINT**
> McNeal: **re-awakening of embryonic induction** in ageing transition-zone stroma may initiate new glands, one theory of BPH.

- Gestational diethylstilboestrol, phthalates and dioxin disrupt development; dioxin-induced hyperinnervation may explain variable α-blocker response.

2 · Zonal anatomy (McNeal)

P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — zones (PZ 70% glands/70% cancers, CZ 25%, TZ 5%/BPH, AFMS), 35° urethral bend, surgical capsule, fasciae, NVB at 5/7 o'clock, urethral/capsular arteries, Santorini plexus, lymph nodes

Not in the paper answer:

- The urethra bends **≈35°** at the verumontanum, where 13–15 ducts open. This is the distal limit of transurethral resection (TURP) and enucleation.
- The transition zone (TZ) has the shortest ducts and gives †≈20% of cancers; the peripheral zone (PZ) has the longest.

> **KEY POINT**
> Trap: the central zone (CZ), around the ejaculatory ducts, is not the TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). The enlarging TZ compresses PZ into the **surgical (pseudo)capsule**, the enucleation plane (see Ch 152).

3 · Fascia, vessels †

- †The capsule is incomplete anteriorly and apically. The neurovascular bundles lie outside it, so enucleation spares them.
- †Prostatic veins join the vertebral (**Batson**) plexus, a route to spinal metastases.

4 · Histology and cell types

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	Prostate-specific antigen (PSA), NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, prostatic acid phosphatase (PAP)	Almost none

- The urethra and proximal ducts carry club, hillock and neuroendocrine cells; BPH stroma contains prominent lymphocytes.

5 · Secretions and clinical markers

- Prostatic fluid is **37–44%** of semen and acidic (pH ≈6.3); seminal vesicle fluid is the alkaline majority.
- Kallikreins liquefy semen within **20–30 min**; incomplete liquefaction at 60 min is nonliquefaction.
- **Ion trapping**: macrolides concentrate in acidic fluid; inflammation alkalinises it.
- PSA, PSMA and PAP all occur outside the prostate, so "prostate-specific" is a misnomer.

Product	Key facts
PSA (KLK3)	Serine protease (19q13.4); liquefies semen
PSMA	Folate hydrolase; rarely deleted in cancer, hence PSMA-PET and ^{177}Lu-PSMA
PAP	First tumour marker (Huggins and Hodges, 1941) MCQ
Zinc	Highest of any tissue; inhibits mitochondrial aconitase, so citrate is secreted

KEY POINT

Benign cells: high zinc blocks aconitase, so citrate is exported. Cancer: **zinc is lost** and citrate is oxidised, so citrate is low (†the basis of MR spectroscopy).

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

1 · Definitions and the four overlapping entities

Histology, size, obstruction and symptoms differ; treatment depends on which.

P2 Pathophysiology, classification and management of LUTS. — BPH vs BPE vs BPO vs BOO, ICS symptom phases, static/dynamic (α1A) components, capsule/TUIP, bladder hypertrophy then decompensation, one-third keep storage LUTS after surgery

Not in the paper answer:

- †BOOI = PdetQmax − 2Qmax: **>40 obstructed, 20–40 equivocal, <20 unobstructed**.
- "Hypertrophy" is wrong because BPH raises cell **number**, not size; "prostatism" is obsolete (Abrams coined "LUTS") MCQ.

Term	How measured
Benign prostatic hyperplasia (BPH): a histological increase in epithelial and stromal cell number in the periurethral/transition zone (TZ)	Histology
Benign prostatic enlargement (BPE): enlargement caused by BPH	Digital rectal examination (DRE), transrectal ultrasound (TRUS), MRI
Bladder outlet obstruction (BOO): obstruction from any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow (best); flow rate (indirect)
Benign prostatic obstruction (BPO): BOO caused by BPE	Pressure–flow
Lower urinary tract symptoms (LUTS): storage, voiding and post-micturition; not prostate-specific	International Prostate Symptom Score (IPSS)

- The four entities overlap only **partially**; BOO may exist without BPE (for example, a stricture). Detrusor ageing, polyuria, sleep disorders and impaired contractility can all mimic BPH.

> **KEY POINT**
> Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are separate; obstruction is diagnosed only by pressure–flow.

2 · Epidemiology

- At autopsy there is **no BPH under 30**; prevalence is ≈ **50% in the 50s**† and **88% in the 80s**. It is **similar across all ethnic and geographic groups**.
- No clinical definition is globally accepted. BPH is described by **symptoms** (IPSS), **enlargement** and **obstruction**, plus bother. The proposed volume cut-off is **>30 mL**; a community composite is IPSS >7, maximum flow rate (Qmax) <15 mL/s and volume >20 mL.

P3 P1 IPSS score; Methods to estimate prostate size — IPSS items, bands 0–7/8–19/20–35, QoL item, no incontinence/pain items, LURN; DRE underestimates, TRUS ellipsoid, MRI, CT

Not in the paper answer:

- Probability of obstruction by Qmax: <10 mL/s high, 10–15 intermediate, >15 low. Qmax falls almost linearly with age (Olmsted median 20.3 mL/s at 40–44 versus 11.3 at 75–79).
- DRE underestimates size by 25% to ≥50%.

ALGORITHM

1. Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**.
2. Transition zone volume (TZV) ≈ 15 → 25 mL.

3 · Aetiology

KEY POINT

BPH is an **accumulation** disease: volume = proliferation minus death. Established BPH shows equal or reduced replication and raised bcl-2. Androgens act chiefly by **suppressing cell death**, so withdrawal shrinks the gland.

- New glands otherwise form only in the fetus, hence the theory of **embryonic reawakening** of stromal induction.

3.1 Androgens: permissive, not causal

- Prepubertal castrates and men with androgen defects **never develop BPH**; withdrawal gives only **partial** involution. The two established determinants are **age** and **functioning testes**.
- Serum testosterone falls with age, but **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. DHT is **maintained, not elevated**; old "raised DHT" reports used cadaveric controls MCQ.
- Serum testosterone does not predict volume or PSA. Androgens act via growth factors rather than as direct mitogens, and they are anti-inflammatory.

ALGORITHM

1. Testosterone enters the stromal cell.
2. **Type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) converts it to DHT.
3. DHT acts on the stroma (autocrine) or diffuses to the epithelium (paracrine).
4. The DHT–AR complex binds DNA and switches on growth factors.
5. Growth factors drive epithelial survival and growth.

- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% is adrenal. DHT has higher AR affinity and forms a more stable complex.

Feature	Type 1	Type 2
Site	Skin, liver	Prostate **stromal fibroblasts**, some basal cells
5α-reductase deficiency	Normal	**Mutated**
Finasteride / dutasteride	Not inhibited / inhibited	Inhibited / inhibited
Role	Possibly cancer	Development and BPH growth

- Finasteride shrinks the gland as much as dual inhibition, so type 1 is unimportant in BPH.
- SRD5A2 lies on **chromosome 2p23** MCQ; its polymorphisms link inconsistently to BPH.

3.2 5α-reductase inhibitor (5-ARI) resistance

- 5-ARIs are the **only medical therapy that reduces volume**.
- Resistance is **not** failure to lower DHT but focal lack of luminal atrophy in nodules, or stromal hyperplasia, often linked to inflammation (NF-κB-driven AR variants, SRD5A2 silencing). The proposed definition is unchanged volume and no atrophy despite lower DHT.

3.3 Oestrogens

- In dogs, androgen plus oestradiol produces BPH; oestrogen induces AR and increases stroma. Oestrogen rises with age relative to testosterone, and adipose tissue aromatises.
- **Oestrogen receptor α (ERα) is stromal** and proliferative; **ERβ is epithelial** and antiproliferative. An ERβ agonist failed in a trial for LUTS.

3.4 Cell death; stromal–epithelial interaction

- Apoptosis after castration is active and probably mediated by transforming growth factor-β (TGF-β).
- Hypothesis: a **stromal defect removes a brake** on epithelial proliferation. Periurethral interstitial fibroblasts (carrying type 2 5α-reductase) increase in BPH and expand the TZ; **no drug targets them**.
- Chemokines such as interleukin-8 (IL-8) turn fibroblasts into collagen-producing myofibroblasts, and the resulting fibrosis contributes to LUTS.

3.5 Growth factors

Androgen permits; growth factors signal and are the likelier primary defect. The first prostatic growth factor found was basic fibroblast growth factor (FGF-2) MCQ.

Factor	Source and target	Role in BPH
FGF-2	Stroma (autocrine)	Raised 2–3-fold; induced by TGF-β and hypoxia
FGF-7 (keratinocyte growth factor, KGF)	Made by stroma; receptor **only on epithelium**	Leading stromal mediator of hormonal control
FGF-10	Smooth muscle to epithelium	Raised by androgens
TGF-β1	Stroma	**Inhibits epithelium** but is **mitogenic for fibroblasts**; BPH stroma shows a blunted response
Insulin-like growth factors (IGF-1, -2)	—	Transgenic IGF-1 causes enlargement; serum IGF-1 is unrelated to volume
Others	—	Epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), FGF-17

- **Hypoxia hypothesis**: hypoxia induces HIF-1, which raises FGF-2 and FGF-7, so growing nodules become more hypoxic; vascular disease such as diabetes may add risk.

3.6 Inflammation

- **Activated T cells** secrete growth factors, and cytokines drive the stroma. Autoimmune disease increases BPH, whereas TNF antagonists are associated with less.
- **Inflammation correlates with enlargement and 5-ARI resistance**, but causality is unproven.

3.7 Metabolic syndrome, lifestyle

- Metabolic syndrome causes prostatic inflammation and LUTS via aromatisation, **hyperinsulinaemia driving sympathetic overactivity**, and low HDL.
- **Waist circumference** predicts prostatectomy and LUTS independently of BMI. BMI >35 versus <25 gives an odds ratio (OR) of **3.5** for a prostate ≥40 mL.

Factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (LUTS)	Increases risk
Physical activity; moderate alcohol	Lowers risk (alcohol lowers BPH, **not LUTS**)
Smoking, diet, vasectomy, sexual activity	No consistent effect (**vasectomy is not a risk factor**)
Jewish religion	More prostatectomy, not more BPH (access to care) MCQ

- **Depression** is strongly linked to LUTS, and LUTS severity correlates with sexual dysfunction.
- **Drugs that worsen LUTS**: α-sympathomimetic cold remedies, antidepressants, antihistamines and bronchodilators; anticholinergic and adrenergic drugs can precipitate retention.

3.8 Genetics and others

- In young men with large glands (>37 g), first-degree relatives have ≈ **4-fold** risk of surgery. Inheritance is **autosomal dominant**; ≈ **50%** of men operated on before 60 have heritable disease, versus ≈9% after 60.
- **Familial BPH** shows early onset, a **large gland**, normal androgens and a normal 5-ARI response. Only **dogs and humans** get BPH.

4 · Pathology

- **Every nodule arises in the TZ or the periurethral region**. The TZ lies outside the preprostatic sphincter; the periurethral zone lies inside it, with glands parallel to the urethra.
- The **middle lobe** is **periurethral**, because there is no TZ at the bladder neck MCQ. Patterns are middle, lateral, trilobar and posterior commissural (median bar).

> **KEY POINT**
> Only humans have a **capsule**, transmitting expansion onto the urethra; dogs get BPH but rarely obstruct. Transurethral incision of the prostate (TUIP) relieves obstruction without changing volume. Hence **size does not correlate with obstruction**.

Feature	Periurethral nodules	TZ nodules
Composition	Purely stromal (early), like **embryonic mesenchyme**	Glandular, tall epithelium
Origin	Interstitial fibroblasts	New duct branches forming a new ductal system

- **Two phases**: for ≈20 years nodule number rises; then glandular nodules grow.
- Small glands are mainly stroma and large glands mainly epithelial nodules. This variable stromal:epithelial ratio explains variable drug response, since α-blockers target smooth muscle and 5-ARIs target epithelium.

5 · Pathophysiology of obstruction

Many symptoms come from the bladder, so some persist after surgery.

ALGORITHM

1. Hyperplasia raises urethral resistance (static and dynamic components).
2. The detrusor compensates, at a cost to storage function.
3. Storage LUTS appear, with or without impaired emptying.
4. In a minority, the detrusor decompensates.

- The static component (bulk under the capsule, elastic ECM) is **not** reduced by α-blockade. The dynamic component is smooth muscle tone via **α1a** receptors (the most abundant subtype).
- Other regulators include endothelin and **phosphodiesterases 4 and 5 (PDE4/5)**, which is why PDE5 inhibitors help LUTS.
- Many LUTS arise from **obstruction-induced bladder change**. ≈ **One-third** of men keep voiding dysfunction (mostly storage) after relief.
- Storage changes are detrusor overactivity (DO) and lower compliance; emptying changes are reduced contractility and, in a minority, detrusor failure.

ALGORITHM

1. **Three-stage remodelling**:
2. Smooth muscle hypertrophy.
3. Compensation.
4. Decompensation (underactive bladder).

- Stretch, pressure and **hypoxia/ischaemia** switch the detrusor to a **secretory phenotype** that lays down ECM.
- **Trabeculation** reflects increased detrusor **collagen** and progresses to cellules, diverticula and failure; there is no accepted grading.

KEY POINT

Storage LUTS persist after transurethral resection of the prostate (TURP) because the collagen-rich, overactive detrusor does not reverse, so counsel men with DO before surgery. Acute urinary retention (AUR) is **not** the inevitable end: many men in AUR have an adequate detrusor and a precipitant.

6 · Natural history

Who progresses decides watchful waiting versus treatment.

- **Regression to the mean**: entry thresholds select men at a temporary extreme, so they seem to improve. It is part of every placebo effect.
- Within trials, IPSS, Qmax and volume barely correlate (restriction of range); only PSA and volume still correlate.
- ≈ **40%** improve without treatment (watchful waiting behaves like placebo); active drug gives about twice the placebo improvement.
- **Olmsted County**: IPSS rises ≈ **0.2–0.34 points/year**; Qmax falls ≈ **2.1%/year** (faster if age ≥70 or Qmax <10); volume grows ≈ **0.6 mL/year** (faster with higher PSA or volume). Volume >50 mL gives ≈3.5-fold odds of moderate–severe LUTS.
- A perceptible improvement is ≈ −2 IPSS points at low baseline versus −6 at high baseline.

Predictor of progression	Predicts
Age	Everything; AUR rises linearly from age 40 to 80+
PSA ≥1.4–1.5 ng/mL	Growth, symptoms and flow, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, new LUTS
IPSS ≥8 or high bother	AUR, surgery
Qmax <10–12 mL/s	AUR, flow decline, treatment
Raised post-void residual (PVR)	Treatment failure, AUR

- **MTOPS** (Medical Therapy of Prostatic Symptoms) placebo arm: progression **4.5 per 100 patient-years** (17% at 4 yr), mostly an **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.
- **PLESS** (Proscar Long-term Efficacy and Safety Study) placebo arm: symptoms drifted back to baseline. Surgery was 10% at 4 yr, rising from ≈6% to ≈15% across PSA and volume tertiles.
- **REDUCE** placebo arm: volume 40–80 mL and Qmax <10 each gave a hazard ratio of 1.67 for new LUTS.

> **KEY POINT**
> PSA is a cheap surrogate for volume, the main driver of AUR and surgery; models with >100 predictors barely beat PSA alone.

7 · Complications

These are the absolute indications for surgery.

> **KEY POINT**
> Death, renal failure, stones and infection are now rare. The common events are **AUR and surgery**, predicted by age, symptoms, low flow and a large gland (or high PSA).

7.1 Acute urinary retention

- Precipitants include infection, overdistension, fluid load, alcohol and bed rest; prostatic infarction is **not** proven.

	Spontaneous AUR	Precipitated AUR
Trigger	None	Surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- AUR incidence is ≈ **4.5–6.8/1000 person-years** in the community, 18 in PLESS placebo and 6 in MTOPS. The hazard is constant over time.
- **Age and symptoms multiply risk**: in Olmsted, men in their 70s with IPSS >7 had 34.7/1000 person-years. Relative risks: IPSS ≥8 3.2; Qmax <12 3.9; volume >30 mL 3.0; PSA >1.4 several-fold.

7.2 Other complications

- **Renal**: "**silent prostatism**" is asymptomatic renal failure from BOO; MTOPS had none in >3000 men. †Catheterise first, watch for post-obstructive diuresis, let creatinine stabilise, then operate.
- **Bladder stones** are ≈ **8×** commoner because of stasis; renal stones are not.
- **Haematuria** reflects increased microvessel density; **finasteride** is first-line because it lowers VEGF. †Exclude cancer and stones first.
- **UTI** is rare (MTOPS 0.1/100 patient-years). **Incontinence** is overflow or urge (DO in up to half).
- **Surgery** is a soft end point and AUR a hard one. In Olmsted, moderate–severe symptoms, Qmax <12, volume >30 and PSA ≥1.4 each gave ≈4× treatment.

8 · Landmark trials

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure RR 0.48; 5-yr 10% vs 21%; ≈36% crossed over; delayed TURP smaller gains	Watchful waiting safe; delay may reduce benefit
PLESS	Finasteride vs placebo, 4 yr	Placebo returns to baseline; AUR/surgery rise with PSA/volume	PSA/volume predict progression
MTOPS	Placebo vs doxazosin vs finasteride vs combination	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4; no renal failure	Defined "clinical progression"
Olmsted County Study	Population cohort, 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural history; age, IPSS, Qmax, volume predict AUR

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

1 · Definitions and the diagnostic problem

Not every man with urinary symptoms is obstructed; find the cause first.

- Benign prostatic hyperplasia (BPH) is **histological** proliferation in the **transition zone (TZ)**. It can lead to benign prostatic enlargement (BPE), benign prostatic obstruction (BPO, a form of bladder outlet obstruction, BOO) and lower urinary tract symptoms (LUTS).
- LUTS are multifactorial, including detrusor overactivity (DO) or underactivity (DUA), stricture and infection (UTI).

> **KEY POINT**
> Bothersome LUTS are not BPH-related BOO until assessed. Aims: **grade severity and bother** and **identify the cause**. Symptoms, flow, post-void residual (PVR), volume and cystoscopy cannot separate obstruction from bladder dysfunction; only the **pressure–flow study** can (gold standard, reserved for when it changes surgery).

Guidelines: American Urological Association (AUA) BPH (2021); European Association of Urology (EAU) non-neurogenic male LUTS (2022).

2 · History and symptom assessment

- **Comorbidities**: diabetes mellitus or insipidus, renal disease, heart failure, sleep apnoea, neurological disease, haematuria and prior lower-tract surgery.
- **Drugs**: loop diuretics, β-blockers, excess fluid and caffeine worsen storage; anticholinergics impair contractility.
- Assess **sexual function**. LUTS do **not** raise prostate cancer (PCa) risk.
- Storage symptoms (frequency, urgency, urge incontinence, nocturia) form **overactive bladder (OAB)**.

P3 IPSS score — AUASI 7 items + QoL (0–6, WHO 1993), subscores not validated, does not diagnose BOO, VPSS/BII/OAB-q alternatives

Not in the paper answer:

- The AUA Symptom Index (AUASI) and International Prostate Symptom Score (IPSS) total **0–35**: mild **0–7**, moderate **8–19**, severe **20–35**.
- The minimum perceptible change is **3 points**; with a severe baseline, **percentage** reduction is more reliable.

3 · Examination and DRE

- Look for a palpable bladder, meatal stenosis and phimosis, and check **anal tone** and lower-limb neurology.
- **Digital rectal examination (DRE)** estimates volume and screens for PCa. It **underestimates prostate volume (PV) above 30 mL**; a flat gland suggests <30 mL.
- PV does not correlate with symptoms or obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · Frequency–volume charts and nocturia

- A frequency–volume chart (FVC) records times and volumes; a bladder diary adds intake and urgency. **Indication**: predominant **storage LUTS or nocturia**. **Duration**: **3 days**, since adherence falls by 7 days.
- **Nocturia** is waking ≥1 time to void. **Nocturnal polyuria** is night output **>20%** of 24-h output (young) or **>33%** (over 65).

ALGORITHM

1. **Nocturia**
2. Complete an FVC.
3. If †global polyuria (>40 mL/kg/24 h), look for diabetes or excess intake.
4. If nocturnal polyuria, look for heart failure, sleep apnoea, evening fluids or diuretic timing.
5. If reduced capacity, think of OAB or BOO.
6. Treat the cause. If nocturia persists, consider **sleep apnoea** (AUA).

Nocturnal polyuria: no fluid 2 h before bed, no evening alcohol or caffeine, retimed diuretics, with or without **desmopressin**. †It risks **hyponatraemia**, so check sodium at baseline, 1 week and 1 month.

5 · Laboratory tests

- **Urinalysis is mandatory** for UTI, microhaematuria, glycosuria and proteinuria. Gross haematuria needs exclusion of bladder cancer; **cytology** is for severe storage symptoms with dysuria, or smokers.
- **Prostate-specific antigen (PSA)** gives PCa risk, a volume surrogate and progression risk. Test only if life expectancy is **>10 years** and it changes management.
- PSA predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s). Baseline PSA also predicts acute urinary retention (AUR), surgery, and change in symptoms and maximum flow rate (Qmax).
- **5α-reductase inhibitors (5-ARIs) lower PSA by 40–50% by 12 months**, so take a baseline first. †Double the PSA after ≥6 months; any rise from the nadir needs evaluation.
- **Creatinine and eGFR** are checked (EAU) if renal impairment or retention is suspected.

6 · Instrumental investigations

PVR

- A PVR **<30 mL** is insignificant; persistently **>50 mL** is important; **200–300 mL** may mean DUA and a poorer response; **>300 mL** is significant chronic retention.
- High PVR means BOO **or DUA**; it predicts symptom deterioration (MTOPS >39 mL; ALTESS >93 mL), **not surgery**.

Uroflowmetry

- Record **Qmax**, voided volume and pattern. Repeat if voided volume is **<150 mL**.
- It **cannot distinguish BOO from DUA** (DUA in 9–48%). Qmax **10 mL/s** is more specific; **15 mL/s** is more sensitive.
- Optional at first; **recommended before active treatment** and for monitoring.

Urodynamics

P2 Role of UDS before prostate surgery — cystometry/PFS, BOOI and BCI, BOO = high pressure + low flow vs DUA, EAU indications, UPSTREAM: not routine

Not in the paper answer:

- Pressure–flow study (PFS) indications (EAU): failed invasive treatment; cannot void **>150 mL**; PVR **>300 mL**; age **>80** or **<50 years** with voiding LUTS; neurological disease or pelvic surgery. AUA: before surgery if **Qmax >10 mL/s**.
- DUA predicts long-term surgical failure.

Imaging and endoscopy

P1 Methods to estimate prostate size — TAUS/TRUS ellipsoid, planimetry, MRI, CT; IPP grading

Not in the paper answer:

- **Intravesical prostatic protrusion (IPP) ≥10 mm** is high grade; it predicts need for therapy and a **failed trial without catheter (TWOC)**. Bladder wall thickness and other non-invasive tests are accurate, but **none replaces PFS**.
- **Transrectal ultrasound (TRUS)** is the method of choice for PV, needed before a 5-ARI and **any BPH surgery**; a median lobe guides the choice of minimally invasive therapy.
- **Cystoscopy** neither diagnoses BOO nor decides treatment. Indications include haematuria, recurrent UTI, suspected stricture, and **excluding a median lobe before prostatic urethral lift (PUL)**.

7 · Evaluation algorithms

ALGORITHM

1. **EAU (men ≥40 yr)**
2. History (including sexual), symptom score, examination with DRE, urinalysis, PVR, and PSA if life expectancy >10 yr.
3. If DRE, PSA or urinalysis is abnormal, or neurological disease, follow the relevant guideline.
4. If not bothersome, no further work-up.
5. If PVR is significant, renal ultrasound with or without renal function.
6. If bothersome, FVC (storage or nocturia), then prostate ultrasound and uroflow if treating, then the medical algorithm.
7. Before surgery, endoscopy if it alters the choice; PFS for specific indications.

ALGORITHM

1. **AUA**
2. History, examination, IPSS, urinalysis, with or without PVR and flow.
3. If **PVR >300 mL**, manage as non-neurogenic chronic retention.
4. Counsel: behavioural, medical, procedural.
5. Reassess at **4–12 weeks**. If improved, follow up. If nocturia predominates, FVC and a drug. If incomplete, FVC, PVR and flow, urodynamics and cystoscopy, then the OAB pathway, surgery, or both.

8 · Watchful waiting (WW)

- **Indication**: mild–moderate uncomplicated LUTS, minimal bother. **Excluded**: gross haematuria, UTI, raised PSA or PVR, and neurological disease.
- WW is reassurance, education, lifestyle and **yearly** review; in the VA trial ≈a quarter crossed over because of **bother**.
- Lifestyle: fluids **1.5–2 L/day**, none 2 h before bed; less caffeine and evening alcohol; retime diuretics; **double voiding** and **urethral milking**.

KEY POINT

Treatment is driven by **bother and complicating features**, not by score or size alone.

9 · Medical therapy: choice

- Drugs are first line without an **absolute surgical indication** (refractory or recurrent AUR, recurrent UTI, renal insufficiency, stones, recurrent haematuria; see Ch 151).

P2 BPH: medical management and surgical modalities — drug classes and doses, MTOPS/CombAT/SMART, IFIS, ALFAUR, desmopressin, surgical options by size

Not in the paper answer:

- Antimuscarinics need PVR <200 mL; avoid oxybutynin in the elderly; hexane-extract Serenoa is an EAU option.

ALGORITHM

1. **EAU**
2. Not bothersome: WW.
3. Nocturnal polyuria: lifestyle with or without desmopressin.
4. Storage predominant: antimuscarinic or β3-agonist.
5. If **PV >40 mL and long-term treatment** intended: 5-ARI with or without an α1-blocker or phosphodiesterase-5 inhibitor (PDE5I).
6. Otherwise α1-blocker or PDE5I; if storage symptoms remain, add an antimuscarinic or β3-agonist. **AUA**: α-blocker (PDE5I if erectile dysfunction, ED) → **prostate >30 mL** → add 5-ARI → PDE5I trial → surgery.

KEY POINT

Reading trials: placebo improves scores and flow, so judge by a 3-point or percentage threshold; the goal includes **preventing progression**.

10 · α1-Blockers

P2 Tamsulosin; Silodosin. — α1a ≈70% of prostatic α1, subtype table, dynamic component, onset, ejaculatory dysfunction, IFIS, silodosin renal dosing, greatest BOOI fall

Not in the paper answer:

- **No α1-blocker is clearly more effective**; they differ in adverse effects (AEs). Doxazosin cut progression 39% (MTOPS), **not AUR or surgery**.
- **ALLHAT**: the doxazosin arm stopped for heart failure, so never use it as sole antihypertensive; **never use two α-blockers**.

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; abandoned

Class	Drug	Dose	Notes
Short-acting	Prazosin, indoramin	bd	Not used for LUTS
Long-acting	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin immediate release (IR)	2–8 mg od	Start 2 mg at bedtime, titrate
	Doxazosin GITS (sustained release)	4–8 mg od	Start **4 mg**, **no titration**
Uroselective	Alfuzosin extended release (ER)	10 mg od	Clinically uroselective
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d
	Silodosin	8 mg od	Most α1a-selective
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia

KEY POINT

Titration: α1b blockade causes **first-dose hypotension** and syncope; a bedtime start and titration avoid it. Uroselective drugs are fixed-dose; doxazosin GITS starts at 4 mg without titration.

- They double the placebo symptom gain with **durable** benefit but do **not** shrink the gland; tamsulosin plateaus at **0.4 mg**.
- **AEs**: dizziness, asthenia, orthostatic hypotension and anejaculation. In the **elderly**, dizziness causes falls; severe hypotension is mostly in the **first 8 weeks**.
- **Sexual**: ED equals placebo. Ejaculatory dysfunction is mostly **anejaculation** (emission failure): silodosin (odds ratio ≈32) > tamsulosin (≈9) > alfuzosin.

KEY POINT

Intraoperative floppy iris syndrome (IFIS): **billowing iris, iris prolapse, progressive miosis**. α1a blockade causes dilator atrophy, so IFIS occurs **years after stopping**; stopping pre-op does **not** help. Highest with **tamsulosin**, then alfuzosin, doxazosin and naftopidil. †If cataract surgery is awaited, defer the α1-blocker.

Drug	Vascular	Ejaculatory	IFIS
Terazosin	High	Low	Reported
Doxazosin	High (GITS lower)	Low	Lower
Alfuzosin	Intermediate	Lowest uroselective	Intermediate
Tamsulosin	Low	Dose-dependent	**Highest**
Silodosin	Lowest	**Highest**	Not well studied

KEY POINT

One efficacy, three toxicities: **α1b blockade gives vascular effects** (terazosin, doxazosin); **α1a blockade affects ejaculation and causes IFIS** (silodosin, tamsulosin); **α1d blockade gives storage benefit** (naftopidil).

11 · 5α-Reductase inhibitors

The only drugs that prevent retention and surgery.

P2 Dutasteride. — type 1+2, DHT ↓90%, 0.5 mg, PV ↓25%, AUR ↓57%/surgery ↓48%, PSA doubling, REDUCE, teratogenic

Not in the paper answer:

- **Finasteride 5 mg** (type 2) lowers dihydrotestosterone (DHT) by †~70% and PV by 18–20%; in PLESS, AUR fell 57% and surgery 55%. It performs the same as dutasteride head to head.
- Indication: moderate–severe LUTS with PV >40 mL (EAU) or >30 mL (AUA).
- They relieve the **static component**. Benefit is **slow** (maximum shrinkage at **6 months**) and **volume dependent**, best at **>40 mL**.
- They are the **only class altering natural history**.
- PCPT and REDUCE showed fewer cancers but slightly more Gleason ≥8, so they are **not licensed for prevention**.
- **AEs**: ED, lower libido and lower ejaculate volume (≈2× placebo, waning after 1–2 yr), **depression** (FDA 2011) and rarely gynaecomastia. **Post-finasteride syndrome** is persistence of symptoms after stopping.

> **KEY POINT**
> 5-ARIs treat the gland: slow, volume-dependent and modest for symptoms, but they prevent AUR and surgery, and they halve PSA (both a trap and a tool).

12 · Antimuscarinics

- They block detrusor **M2/M3** receptors, mainly on **afferent** pathways.
- **Indication**: predominant storage LUTS, **PVR <200 mL** and no prior AUR.
- Agents: darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine ER 4 mg and trospium.
- In BOO, PVR rises slightly and AUR equals placebo (trials excluded PVR ≥200).
- AEs: **dry mouth**, constipation and cognitive impairment. **Oxybutynin** has most brain penetration, so avoid it in the elderly; trospium least†.

> **KEY POINT**
> Antimuscarinics treat the bladder, so every recommendation is anchored to a PVR ceiling (200 mL).

13 · β3-Agonists

- β3 stimulation gives **storage relaxation without weakening voiding** (Qmax unchanged), with less dry mouth than antimuscarinics.
- **Mirabegron** starts at **25 mg** and rises to **50 mg**; in BOO, PVR rose only at 100 mg.
- AEs: nasopharyngitis, headache, blood pressure rise ≤1 mm Hg. **Contraindicated** in uncontrolled hypertension **≥180/110 mm Hg**; it is a moderate CYP2D6 inhibitor†.
- **Vibegron** 75 mg has no CYP2D6 inhibition; data are mainly in women.

14 · PDE5 inhibitors

- **Tadalafil 5 mg daily** is the only licensed agent (2.5 mg is ineffective). It lowers IPSS by ≈6 (placebo 3.6), **independent of ED**, and reduces nocturia. **Qmax is unchanged**.
- First choice for LUTS with ED (AUA). AEs: headache, back pain, dyspepsia, flushing. †Contraindicated with nitrates and riociguat.

15 · Combination therapy

- **MTOPS** (Medical Therapy of Prostatic Symptoms): progression = AUASI rise **≥4**, AUR, UTI or urosepsis, incontinence, or creatinine rise ≥1.5 mg/dL. Placebo 17%. Number needed to treat (NNT) for combination 8.4 (≈5 if PSA >4 and PV >40).
- **CombAT** (PV ≥30, PSA ≥1.5): AUR or surgery fell two-thirds versus tamsulosin but not significantly (NS) versus dutasteride.
- **Indication**: high risk, PV **>30–40 mL** and PSA **>1.5 ng/mL**. **SMART**: stop the α-blocker after 6 months.

MTOPS reduction vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

KEY POINT

α-blockers prevent symptomatic progression only; 5-ARIs prevent AUR and surgery.
The α-blocker gives fast dynamic relief; the 5-ARI gives slow static shrinkage.

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS (tolterodine 4-arm; NEPTUNE)	PVR <150–200 mL only; AUR <1–3%
α1-blocker + β3-agonist	Persistent storage (MATCH)	PVR rise; limited data
α1-blocker + PDE5I	LUTS with or without ED; small gain	Hypotension; not recommended; start lowest dose
PDE5I + 5-ARI	Large prostate with or without ED	Single trial; FDA capsule 2021

16 · Phytotherapy

- Evidence is product-specific. Most **Serenoa repens** extracts show no benefit, but the **hexane extract (Permixon) 320 mg daily with food** reduced nocturia and raised Qmax and is an EAU option. AUA/EAU make **no general recommendation** for phytotherapy.

17 · Follow-up

- Worsening symptoms and rising PVR predict AUR.

ALGORITHM

1. Review at **4–12 weeks** (α-blocker response, AEs).
2. Review at **6 months** (5-ARI).
3. Then **annually**: score, flow, PVR; PSA on a 5-ARI if life expectancy >10 yr. WW: yearly.

18 · Acute urinary retention

P2 Catheter free trial — spontaneous vs precipitated AUR, α-blocker before TWOC (ALFAUR), predictors of failure, recurrence, 5-ARI prevention

Not in the paper answer:

- CombAT: AUR fell 68% versus tamsulosin but only 18% (NS) versus dutasteride; α-blockers do not prevent AUR (ALTESS, MTOPS).
- About two-thirds of AUR is spontaneous, with a worse TWOC outcome than precipitated AUR.

ALGORITHM

1. Catheterise (†suprapubic if urethral fails).
2. Start an **α1-blocker**.
3. Do a **TWOC at ~3–5 days**.
4. If he voids, continue (add a 5-ARI for a large gland).
5. If he fails, try up to **3 TWOCs**.
6. If AUR recurs or TWOC fails: PFS and surgery (see Ch 151).

- TWOC success is ≈ **60%**, and recurrence is mostly within **6 months**.
- Failure predictors: age, severe LUTS, drained volume **>1 L**, spontaneous AUR and high-grade IPP. 5-ARIs prevent AUR (PLESS, MTOPS).

KEY POINT

The α-blocker rescues the TWOC (dynamic component); **the 5-ARI prevents the next episode** (static component).

19 · Landmark trials

Trial	Design	Result	Take-home
VA transurethral resection (TURP) vs WW	Randomised trial, moderate LUTS	TURP better; most WW did not fail; crossover by bother	WW safe
PLESS	Finasteride vs placebo, 4 yr, PV ~55 mL	AUR 7% to 3%, surgery 10% to 5%; combined fall 51%	5-ARI prevents AUR/surgery
Dutasteride phase III	2 yr, PV ≥30 mL	DHT fell 90%, PV 26%, AUR 57%, surgery 48%	≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 yr	Progression fell 39/34/66%; AUR/surgery only with 5-ARI	Combination for high risk
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 yr	AUR/surgery fell 66% vs tamsulosin	Combination for large gland
SMART	α-blocker withdrawal	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI prevention	Fewer cancers, more high grade	Not licensed
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

1 · Scope and principles

- **American Urological Association (AUA) 2023 operations**: transurethral incision (TUIP), resection (TURP) and vaporisation (TUVP) of the prostate; photoselective vaporisation (PVP); prostatic urethral lift (PUL; UroLift); water vapour thermal therapy (WVTT; Rezūm); temporary implanted prostatic device (TIPD; iTind); robotic waterjet treatment (RWT; Aquablation); holmium or thulium laser enucleation (HoLEP/ThuLEP); simple prostatectomy; prostate artery embolisation (PAE).
- Transurethral microwave therapy (TUMT) was removed in 2023. Because the sham effect is large, new devices need sham-controlled trials.

KEY POINT

Volume predicts progression (acute urinary retention (AUR), surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County).

2 · Work-up and indications for surgery

- History, examination, IPSS and urinalysis at baseline (Statement (St) 1). International Prostate Symptom Score (IPSS) bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery, post-void residual (PVR) is **required** (St 6). **Consider** size and shape by transrectal ultrasound (TRUS), abdominal ultrasound, cystoscopy, MRI or CT (St 5), uroflowmetry (St 7), and pressure–flow study if the diagnosis is uncertain (St 8).
- A raised prostate-specific antigen (PSA) needs biopsy **before** BPH surgery. Counsel on failure and retreatment (St 9).

AUA Statement 26 (surgery recommended)	† European Association of Urology (EAU): absolute vs relative
Renal insufficiency from BPH; refractory retention; recurrent urinary tract infections (UTIs); recurrent bladder stones or gross haematuria from BPH; lower urinary tract symptoms (LUTS) refractory to, or unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation or renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5α-reductase inhibitor (5-ARI). **Relative**: moderate–severe bothersome LUTS failing or declining medical therapy

- **Antibiotics (AUA 2019)**: one dose suffices for most transurethral cases. Cover gram-negatives and enterococci (1st/2nd-generation cephalosporin or co-trimoxazole), guided by antibiogram and urine culture. **Defer surgery if UTI is active and symptomatic**.

3 · Choosing the operation

Volume / situation	AUA 2023 options	Restrictions
<30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes

Volume / situation	AUA 2023 options	Restrictions
>80–150 cc / >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc
Higher bleeding risk	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Preserve erection and ejaculation	PUL or WVTT (St 37)	Within 30–80 cc

KEY POINT

Only enucleation spans every size band, joined above 80 cc only by simple prostatectomy. Minimally invasive surgical therapies (MISTs) and RWT live in 30–80 cc, so volume and median lobe assessment come first.

4 · Monopolar TURP (m-TURP): the benchmark

Every newer operation is judged against TURP.

P1 P2 TURP instruments and techniques; Measures to decrease complications of TURP — resectoscope, Nesbit/floor-first resection, canoe chips, prevention of each complication

Not in the paper answer:

- AUA: TURP for <80 cc (St 28), monopolar or bipolar by expertise (St 29). Monopolar irrigant is non-ionic (water, **1.5% glycine**†, sorbitol) and **hypo-osmolar**, which causes TUR syndrome.

ALGORITHM

1. Inspect the bladder. If tumour, resect it and defer TURP.
2. Identify orifices, neck, trigone, verumontanum (veru), sphincter; if orifices hidden, give indigo carmine or methylene blue.
3. Resect median lobe, then lateral lobes, then anterior tissue and apex last.
4. Evacuate chips, secure haemostasis, insert a large three-way catheter.

- **Anterior and apex last**, because the anterior fossa perforates easily and the sphincter lies more proximally anteriorly.
- **Arterial bleeding** is bright and must be fulgurated. **Venous bleeding** is darker and vanishes with a full bladder; once arteries are secured, use balloon traction.
- Even routine TURP absorbs **800–1000 mL** irrigant. The catheter comes out at 24–48 h.
- Warn of dysuria for weeks and eschar bleeding at **1–4 weeks**. Results last about 10 years.

5 · TURP complications

Missed TUR syndrome can kill.

P1 P3 TUR Syndrome — mechanism, glycine toxicity, features, risk factors, 3% saline protocol

- Risk factors are a gland **>45 cc**, resection **>90 min**, perforation and irrigant height. Keep the bag at about **60 cm**, because prostatic venous pressure is only about **10 mm Hg**. Na^+ **<120 mEq/L** is severe.

ALGORITHM

1. Stop or expedite surgery, secure haemostasis, stop irrigation; send Na^+.
2. If overloaded, give a †loop diuretic.
3. If severe central nervous system signs: **3% saline 100 mL** (+2–3 mEq/L), repeat ×2 at 10 min (max 300 mL).
4. Target +4–6 mEq/L; ≤12 mEq/L in 24 h (≤8–10 if presentation >48 h). Monitor in high dependency.

KEY POINT

Acute post-TURP hyponatraemia is safer to correct than chronic: the brain has not yet shed its organic osmolytes, so osmotic demyelination is less likely. Bipolar saline TURP eliminates TUR syndrome.

- **Perforation** shows glistening fat. A limited extraperitoneal tear needs only a longer catheter, an extensive one needs drainage, and an intraperitoneal tear needs laparotomy. A persistent intra-operative erection responds to intracorporeal phenylephrine.

Late complication	Cause / features	Management
Bladder neck contracture (BNC)	Over-resection or over-fulguration (small glands); onset ~6 months, flow declines	Dilate, then incise over a wire into perivesical fat. Recurrent: add mitomycin C (10 mL of 0.4 mg/mL). Intractable: Y-V plasty
Stricture	Meatus or fossa navicularis	Dilate meatus 2 Fr above sheath; plenty of gel
Retrograde ejaculation	About **two-thirds**	Counsel; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost; apical over-resection causes total stress incontinence (SUI). Rare, usually transient	Veru is the distal limit; †pelvic floor exercises, then †sling or artificial sphincter
Bleeding	Early: incomplete haemostasis. Delayed (1–4 weeks): eschar	Early major: theatre. Delayed mild: fluids; †clot evacuation if retention
Erectile dysfunction (ED)	No conclusive causal evidence	—
Retreatment	Residual apical/anterior tissue, >80 cc, regrowth, BNC, stricture	Rises with size and pre-op retention

6 · Bipolar TURP (b-TURP)

P2 Monopolar v/s Bipolar TURP — saline, no TUR syndrome (NNT ≈50), less clot retention (NNT ≈20), equal efficacy to 5 y

- It is equally effective and safer, so it may replace m-TURP as gold standard.

7 · TUIP

- TUIP releases capsule and bladder neck without removing adenoma. It is for **<30 cc only** (St 31) without a median lobe (relative contraindication; EAU the same). It suits a young man with a small gland who wants ejaculation.

ALGORITHM

1. Use a Collins knife, loop, or holmium/thulium laser.
2. Incise at 5 and/or 7 o'clock from distal to the orifice, through the bladder neck, to proximal to the veru.
3. Deepen to capsule (with or without fat); cauterise.
4. Catheter out and home on day 0–1.

- Compared with TURP it is shorter, bleeds less and relieves symptoms similarly, but gains less Qmax and needs **more reoperation**; retrograde ejaculation is rarer and BNC does not occur.

8 · Bipolar TUVP

- A broad electrode (PlasmaButton) vaporises and coagulates together, so haemostasis is excellent, but there is **no specimen**. Only **bipolar** is recommended (St 32: <30 and 30–80 cc); monopolar was abandoned for irritation, SUI and poor durability.
- Compared with TURP it needs a shorter catheter and fewer transfusions, but causes more dysuria, AUR and retreatment; retrograde ejaculation is as common.

KEY POINT

Vaporisation trades haemostasis for a coagulated, specimen-free bed. The coagulum causes dysuria and retention; with no anatomical plane, durability is weaker. PVP makes the same trade.

9 · Laser physics

P3 Lasers in BPH, including HoLEP — wavelength/chromophore/penetration table (thulium **0.2 mm** shallowest; Nd:YAG abandoned)

Not in the paper answer:

- Deeper penetration leaves more coagulum, hence Nd:YAG's sloughing and retention.

10 · PVP (GreenLight)

PVP suits anticoagulated men with glands under 80 cc.

P4 Green light lasers — physics, technique, AUA <80 cc + bleeding risk, GOLIATH, complications

Not in the paper answer:

- AUA St 33 recommends 120 W or 180 W PVP for <30 and 30–80 cc; anticoagulated men improve equally without more transfusion.

11 · HoLEP

Size-independent, and the only endoscopic operation beating TURP.

P4 P1 HoLEP — size-independent (St 38/43), only endoscopic method superior to TURP; settings, steps, learning curve 20/50, BNC if <40 cc

Not in the paper answer:

- It is durable to 18 years because the whole transition zone (TZ) goes, whereas TURP leaves a peripheral rim.

ALGORITHM

1. Incise the bladder neck at 5 and 7 o'clock (at 6 o'clock if no median lobe); carry to the veru and enucleate the median lobe.
2. Make a perimontanal inverted-U incision; open the apical plane.
3. Sweep anteriorly to 11–1 o'clock; incise the apical mucosal strip (no sphincter traction).
4. Develop the anterior plane to the neck ("Baleen" sign = bladder); release lobes into the bladder.
5. Haemostasis with a defocused beam; morcellate via nephroscope; 22 Fr three-way catheter, ~60 mL balloon.

- The smooth reflective capsule is the depth reference. If the morcellator engages mucosa, stop cutting and suction.
- **Transient SUI** is common early, in proportion to volume removed; early apical release reduces it.
- Retrograde ejaculation occurs in about **three-quarters**, but median-lobe-only HoLEP preserves it in ~90%. Erections are unchanged or improved and retreatment is rare.

12 · Thulium (ThuLEP) and variants

- Variants include ThuVARP (vaporesection), ThuVEP (vapo-incision) and ThuLEP (blunt dissection); steps match HoLEP.
- Function about equals HoLEP, with less bleeding than TURP (St 38 and 43). Retrograde ejaculation is ~four-fifths; erections are preserved.

	HoLEP	ThuLEP / ThuVEP
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MISTs

The ideal MIST is office-based, single session, needs no general anaesthesia, spares sexual function and ends medication. **Trade-off**: retreatment within 5 years exceeds surgery, and durability beyond 5 years is uncertain.

P4 UroLift (prostatic urethral lift) — implant, 30–80 mL no median lobe, >1.5 cm from neck, L.I.F.T./BPH6, encrustation

Not in the paper answer:

- AUA keeps 30–80 cc and has not adopted the FDA ≤100 cc label.
- PUL typically uses **4–6 implants**, placed anterolaterally between dorsal veins and neurovascular bundles.
- **WVTT (Rezūm)** is **convective**: radiofrequency steam is injected into the TZ, and condensation releases latent heat, causing 1.5–2 cm of necrosis confined by the capsule.
- Injections are given every 1 cm from neck to veru, with extra ones for a **median lobe**, under sedation or prostatic block.
- **Rezūm II** showed ~50% IPSS gain to 5 years, retreatment 4.4% and **no de novo ED**; reduced ejaculate settles by 12 months. (St 36/37).

KEY POINT

PUL moves tissue; WVTT ablates it. So PUL cannot treat a median lobe but needs no catheter and causes little dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritation. Both preserve the bladder neck, and so antegrade ejaculation.

14 · Robotic waterjet (RWT, Aquablation)

- A non-thermal saline waterjet ablates tissue mapped on real-time biplanar TRUS. It **needs general or spinal anaesthesia, so it is not a MIST**. It suits 30–80 cc (St 39), treats median lobes and gives no specimen.
- **Haemostasis is the weakness**: traction alone led to bleeding and transfusion, so **focal bladder-neck cautery** via resectoscope is now used.
- **WATER** showed non-inferiority to TURP to 5 years, with retrograde ejaculation 10% vs 36%.

15 · Temporary devices

- **iTind (TIPD)** is a nitinol device left **5–7 days** to cause ischaemic remodelling, then removed. St 41: 25–75 cc, no obstructing median lobe. In its sham trial the responder end point was met, but the absolute IPSS difference was not significant.
- **Optilume** (paclitaxel balloon) has single-arm data only and is not on the AUA list.

16 · Prostate artery embolisation (PAE)

No surgery, but less effective.

ALGORITHM

1. Femoral access under local anaesthesia; pelvic angiography with the catheter balloon as landmark.
2. Pass a 1.9–2.4 Fr microcatheter into the prostatic arteries (mostly from internal pudendal; often duplicated).
3. Inject microspheres to stasis; bilateral is better.

- **Contraindications**: underactive, acontractile or neurogenic bladder; stricture or BNC; large diverticula or stones; contrast allergy; chronic kidney disease.
- AUA St 40: by trained clinicians; no size limit; a role in refractory BPH haematuria. EAU: for men accepting results inferior to TURP.
- Compared with TURP, IPSS, Qmax and volume reduction are inferior but the stay is shorter; most men reintervene by 10 years.
- Complications include non-target embolisation, post-embolisation syndrome and AUR.

17 · Overall comparison

This is the counselling sheet.

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/general (GA); 1–2 d	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
TUIP	<30 cc	Day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritation	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT

The "gold standard" for a given man is the operation that meets his goals safely: a shared decision on symptoms, size and shape, bleeding risk, ejaculation wishes and surgeon expertise.

18 · Landmark trials

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	Lesser-efficacy option

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

1 · Definition and surgical anatomy

- It enucleates only the benign prostatic hyperplasia (BPH) adenoma and is the gold standard for outlet obstruction with glands **>80 g**.
- The plane is the **surgical capsule (pseudocapsule)**: peripheral zone compressed by the enlarging transition zone (see Ch 148).
- **Urethral arteries enter posterolaterally just distal to the bladder neck**, so a figure-of-eight suture goes at **5 and 7 o'clock**.
- Eponyms MCQ : **Freyer** is suprapubic transvesical and **Millin (1945)** is retropubic.

2 · Indications

- Gland **>80–100 g** with bothersome lower urinary tract symptoms (LUTS) refractory to medical or endoscopic therapy.
- Retention, recurrent urinary tract infection (UTI), **bladder stones**, haematuria or obstructive nephropathy; a diverticulum needing surgery (relative); or inability to lie in lithotomy for transurethral resection (TURP).

3 · Pre-operative evaluation

- Check bleeding history, anticoagulants, prior pelvic surgery (which dictates the route), **urine culture** (treat first) and coagulation.

> **KEY POINT**
> Work-up asks three questions. Is the gland big enough (transrectal ultrasound)? Is there cancer (digital rectal examination, prostate-specific antigen, MRI)? Will the bladder empty (urodynamics)?

4 · Open techniques

Freyer opens the bladder; Millin opens the capsule.

P1 P2 Millin's prostatectomy — Retzius, transverse capsulotomy ~2 cm distal to bladder neck, bladder not opened, apex divided under vision, 5/7 o'clock sutures, watertight closure; Millin vs Freyer table

Not in the paper answer:

- **Freyer** uses a transverse **cystotomy 1–2 cm proximal to the bladder neck**. It treats **median lobe, diverticula and stones**, but the apex is blind.
- **Millin**'s drawback is **dorsal venous complex (DVC) bleeding**.

> **KEY POINT**
> Every open variant: enter, find the pseudocapsular plane, divide the urethra at the apex **proximal to the verumontanum** without pulling on the sphincter, then 5/7 o'clock haemostasis with bladder neck mucosal advancement.

5 · Haemostasis

Technique	When / how
Pack fossa 5 min	First
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule
Bladder neck mucosal advancement ("trigonisation")	Haemostasis; prevents contracture
11 and 1 o'clock sutures	Anterior branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC and lateral pedicles, sparing neurovascular bundles (NVBs)
O'Conor plication	Diffuse bleeding: plicate the posterior capsule
Malament purse-string	Persistent bleeding: purse-string round the neck, tied suprapubically, partitions fossa from bladder

ALGORITHM

1. Catheter traction (50 mL balloon), with or without continuous bladder irrigation (CBI); exclude coagulopathy.
2. If bleeding continues, evacuate clot endoscopically and fulgurate.
3. If severe, reopen the fossa.

6 · Postoperative care (open)

- On day 2, if urine is clear, remove the Foley for a trial of void. If drain output is high, check **drain creatinine**.

7 · Laparoscopic and robotic

P3 Laparoscopic Trans vesical Prostatectomy — steps, same function as open with less morbidity

Not in the paper answer:

- Mucosal advancement usually means **no CBI, drain or suprapubic tube**, and the catheter comes out on day 5–7.
- Avoid posterolateral cautery, because thermal NVB injury causes erectile dysfunction (ED).

ALGORITHM

1. Vertical cystotomy; identify orifices.
2. Incise the bladder neck 3→9 o'clock under the median lobe; develop the **posterior plane** to the apex, then the lobes.
3. Divide the anterior commissure, see the veru, and transect the urethra **just proximal to the veru**.
4. Place 5/7 barbed sutures; do **vesicourethral mucosal advancement**; close in two layers.

- **Single-port transvesical surgery** with **pneumovesicum 8 mm Hg** allows same-day discharge.

8 · Outcomes and comparisons

The route changes recovery, not function.

Comparison	Bottom line
Robotic vs open	Less bleeding, shorter catheter and stay, fewer complications; longer operation. Function **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, **no TUR syndrome**, treats stones and diverticula; but more bleeding, longer stay
†vs holmium laser enucleation (HoLEP)	†Equivalent outcomes; HoLEP bleeds less, but has a long learning curve and cannot treat diverticula

KEY POINT

Minimally invasive routes improve only perioperative morbidity; function is identical because enucleation is identical. Open transfusion is now **<5%**.

9 · Complications

Complication	Cause / rate	Management
Extravasation	Incomplete closure; capsular tear	If drain creatinine equals urine, keep drain and Foley
Ureteric orifice injury	Neck incision or sutures	Stent
Bladder neck contracture	Open **2–6%** ; minimally invasive <1%	Dilation or Collings knife incision
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare	Bulking (mild); artificial sphincter (severe)
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op
UTI, stricture, venous thromboembolism (VTE)	VTE <1%	Prophylaxis

KEY POINT

The adenoma lies inside the pseudocapsule and proximal to the veru. Removal loses the bladder neck (retrograde ejaculation) but spares sphincter and NVBs, so incontinence and ED are rare.

★ Putting it together: the BPH pathway

A · Man with LUTS

ALGORITHM

1. **1. Assess:** International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), rectal examination, urinalysis, PSA if it changes management; uroflow and post-void residual (PVR) before intervention. **2. If complications, operate:** refractory retention, recurrent UTI, stones, prostatic haematuria, renal impairment. **3. If mild**, watchful waiting. **4. If bothersome**, drugs (B); if they fail, a procedure (C).

B · Medical therapy

Drug	Use	Notes
α1-blocker	First line; works in days	Ejaculatory dysfunction; floppy iris
5α-reductase inhibitor (5-ARI)	Prostate >30–40 mL	Less retention and surgery; halves PSA
α-blocker and 5-ARI	Large gland, progression risk	Best (MTOPS, CombAT)
Add antimuscarinic or β3-agonist	Persistent storage LUTS	Check PVR first
Tadalafil 5 mg daily	LUTS with ED	No nitrates
Desmopressin	Nocturnal polyuria	Check sodium if ≥65

C · Choosing a procedure

Situation	Options
<30 mL	TUIP or TURP
30–80 mL	Bipolar TURP, PVP, HoLEP/ThuLEP, Aquablation; MISTs (urethral lift, water vapour)
>80 mL	HoLEP/ThuLEP; simple prostatectomy (stones/diverticulum)
Anticoagulation continued	HoLEP, ThuLEP, PVP
Preserve ejaculation	Urethral lift (no median lobe) or water vapour
Unfit	Long-term catheter or †prostatic artery embolisation

D · TURP complications

KEY POINT

TUR syndrome causes confusion and hyponatraemia. Stop, check sodium, give a diuretic if overloaded and 3% saline if severe; bipolar prevents it. Late problems are retrograde ejaculation (~two-thirds), stricture, bladder neck contracture and incontinence (see Ch 151).